ECE 340: Semiconductor Devices

Lecture 17: Diffusion and Recombination

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Band Diagrams With Electric Fields

- As the electron is accelerated in the field, it loses potential energy and gains kinetic energy
- When an electric field is present, the bands are tilted to reflect the loss in potential energy as the electron moves in the direction opposite to the field
- The electric field can be externally applied or internally generated by doping gradients

$\mathcal{V}(x) \equiv$ Electrostatic Potential

$E(x) \equiv$ Electron Potential Energy

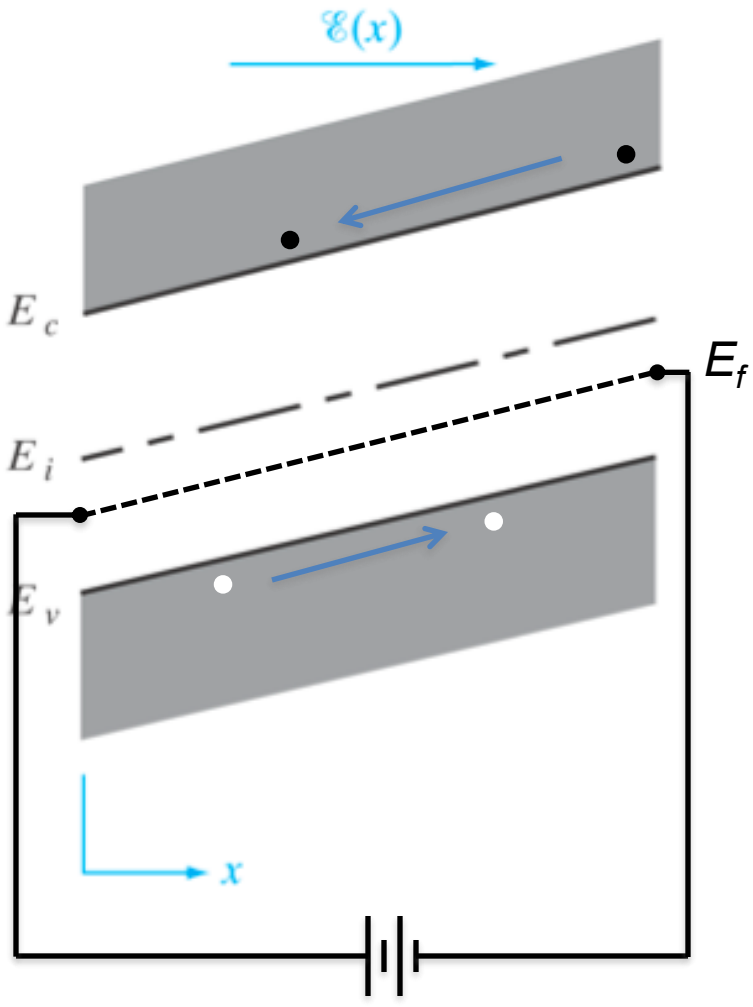
$\mathcal{E}(x) \equiv$ Electric Field

$$\mathcal{V}(x) = \frac{E(x)}{(-q)} \text{ and } \mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx}$$

Choosing E_i as a spatial reference:

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx} = -\frac{d}{dx}\left[\frac{E_i}{(-q)}\right] = \frac{1}{q}\frac{dE_i}{dx}$$

Externally Applied Field:



(Recap) Derivation of Einstein Relationship

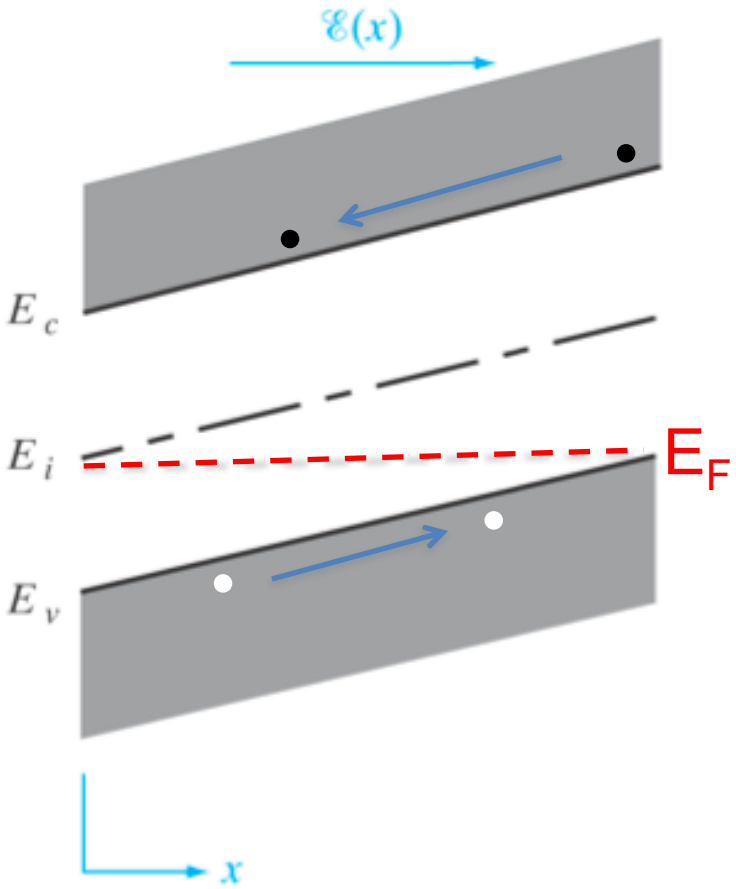
- E field can be generated internally.
- How can we generate the linear field?

$$E_i(x) = E_i(0) + q\mathcal{E}_0 x, \qquad \mathcal{E}_0 = \frac{1}{q}\frac{dE_i}{dx} > 0.$$

$$p_0(x) = n_i e^{|E_i(x) - E_F|/k_B T}$$

- Exponentially increasing doping concentration along *x* can generate linear field.
- Fermi level is invariant (equilibrium) → there is no net current (why?)

Internally Generated Field:



We can also have tilted bands with no externally applied potential!

(Recap) Derivation of Einstein Relationship

$$J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx} = 0$$

$$\text{so } \mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{p(x)}\frac{dp(x)}{dx}$$

$$p_0(x) = n_i e^{(E_i(x)-E_F(x))/kT}.$$

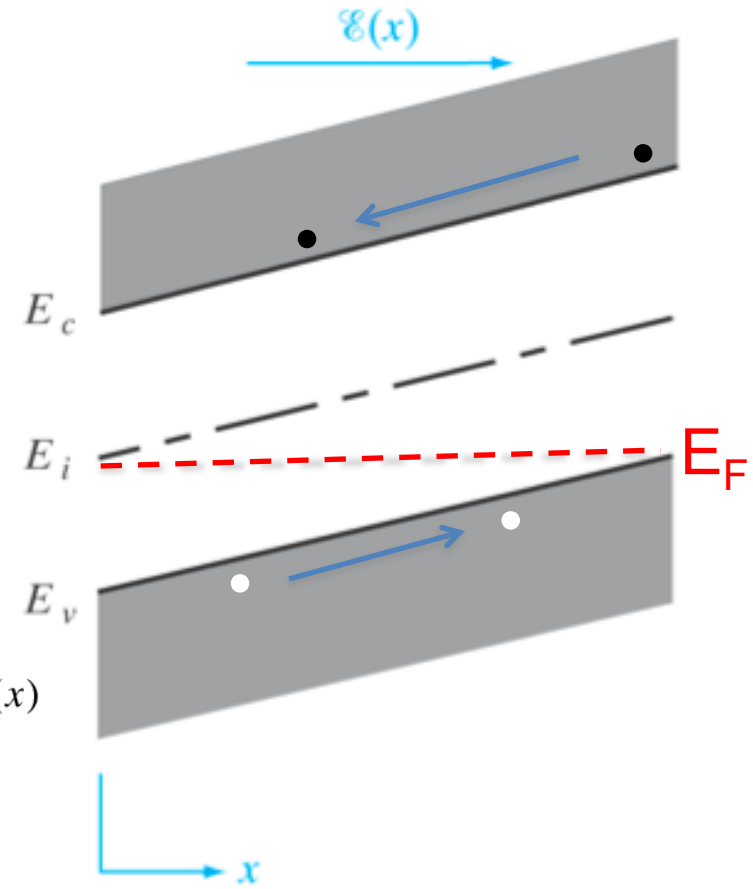
$$\text{using } \frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$$

$$\frac{dp_0}{dx} = n_i e^{(E_i-E_F)/kT}\frac{d}{dx}\left[\frac{E_i - E_F}{kT}\right]$$

$$\mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{kT}\left(\frac{dE_i}{dx} - \frac{dE_F}{dx}\right) = \frac{D_p}{\mu_p}\frac{1}{kT}(q\mathcal{E}(x) - 0) \text{ since } \frac{1}{q}\frac{dE_i}{dx} = \mathcal{E}(x)$$

$$\text{so } 1 = \frac{D_p}{\mu_p}\frac{q}{kT} \text{ and } \boxed{\frac{D_p}{\mu_p} = \frac{kT}{q}}$$

Internally Generated Field:



We can also have tilted bands with no externally applied potential!

(Recap) Electron and Hole Continuity Equations

With Generation considered:

Hole Continuity Equation

$$\left.\frac{dp(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(+q)}\frac{J_p(x)-J_p(x+\Delta x)}{\Delta x} - \frac{\delta p}{\tau_p} + G$$

as $\Delta x \to 0$

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} + G$$

Electron Continuity Equation

$$\left.\frac{dn(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(-q)}\frac{J_n(x)-J_n(x+\Delta x)}{\Delta x} - \frac{\delta n}{\tau_n} + G$$

as $\Delta x \to 0$

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n} + G$$

Continuity Equations

$$\frac{\partial n}{\partial t} = G_n - U_n + \frac{1}{q}\nabla \bullet \vec{J}_n$$

$$\frac{\partial p}{\partial t} = G_p - U_p - \frac{1}{q}\nabla \bullet \vec{J}_p$$

Divergence of a vector field

For a given volume of semiconductor, the rate change of carrier is the net effect of current flow into the volume and generation and recombination rates within the volume.

G_n : *electron generation rate*

G_p : *hole generation rate*

U_n : *electron recombination rate*

U_p : *hole recombination rate*

Diffusion Equation Where Drift is Negligible

Hole Diffusion Equation

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$$

$$\text{and } J_p(x) = -qD_p\frac{dp(x)}{dx} = -qD_p\frac{\partial \delta p}{\partial x} \text{ (diffusion)}$$

$$\text{so } \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial}{\partial x}\left(-qD_p\frac{\partial \delta p}{\partial x}\right) - \frac{\delta p}{\tau_p}$$

$$\therefore \boxed{\frac{\partial \delta p}{\partial t} = D_p\frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}}$$

Electron Diffusion Equation

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$$

$$\text{and } J_n(x) = qD_n\frac{dn(x)}{dx} = qD_n\frac{\partial \delta n}{\partial x} \text{ (diffusion)}$$

$$\text{so } \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial}{\partial x}\left(qD_n\frac{\partial \delta n}{\partial x}\right) - \frac{\delta n}{\tau_n}$$

$$\therefore \boxed{\frac{\partial \delta n}{\partial t} = D_n\frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}}$$

Diffusion Equations

Steady-State Carrier Injection

Diffusion Length

Diffusion Length: Steady State

Steady-State Diffusion Equations

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

In steady state, $\frac{\partial \delta p}{\partial t} = 0$

$$\therefore\ D_p \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{\tau_p} \text{ or } \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{D_p \tau_p} \equiv \frac{\delta p}{L_p^2}$$

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$

In steady state, $\frac{\partial \delta n}{\partial t} = 0$

$$\therefore\ D_n \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{\tau_n} \text{ or } \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2}$$

Hole Injection at x=0



(The injection mechanism is not specified.)

L_p is the average distance a hole travels before recombining.
L_n is the average distance an electron travels before recombining.

$$L_n \equiv \sqrt{D_n \tau_n} \text{ and } L_p \equiv \sqrt{D_p \tau_p}$$

Excess Hole Distribution and Current

Solution to Hole Diffusion Equation

$$\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2} \qquad \frac{\partial \delta p}{\partial t} = 0$$

Solution:

$$\delta p(x) = C_1 e^{x/L_p} + C_2 e^{-x/L_p}$$

Boundary Condition:

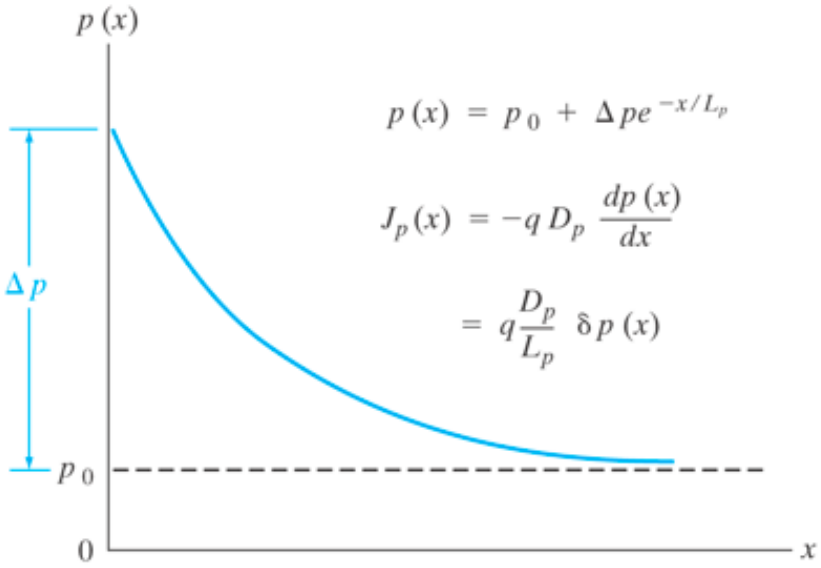
$$\delta p(x=0) = \Delta p = C_1 + C_2$$

$$\delta p(x=\infty) = 0 = C_1 e^{x/L_p}$$

$$C_1 = 0, \quad C_2 = \Delta p$$

$$\delta p(x) = \Delta p \, e^{-x/L_p}$$

Hole Injection at x=0



$$J_p(x) = -qD_p \frac{dp}{dx} = -qD_p \frac{d(p_o + \delta p)}{dx} = -qD_p \frac{d\delta p}{dx}$$

$$= q\frac{D_p}{L_p}\Delta p e^{-x/L_p} = q\frac{D_p}{L_p}\delta p(x)$$

Excess Electron Distribution and Current

Solution to Electron Diffusion Equation

$$\frac{d^2\delta n}{dx^2} = \frac{\delta n}{L_n^2} \qquad \frac{\partial \delta n}{\partial t} = 0$$

Solution:

$$\delta n(x) = C_1 e^{x/L_n} + C_2 e^{-x/L_n}$$

Boundary Condition:

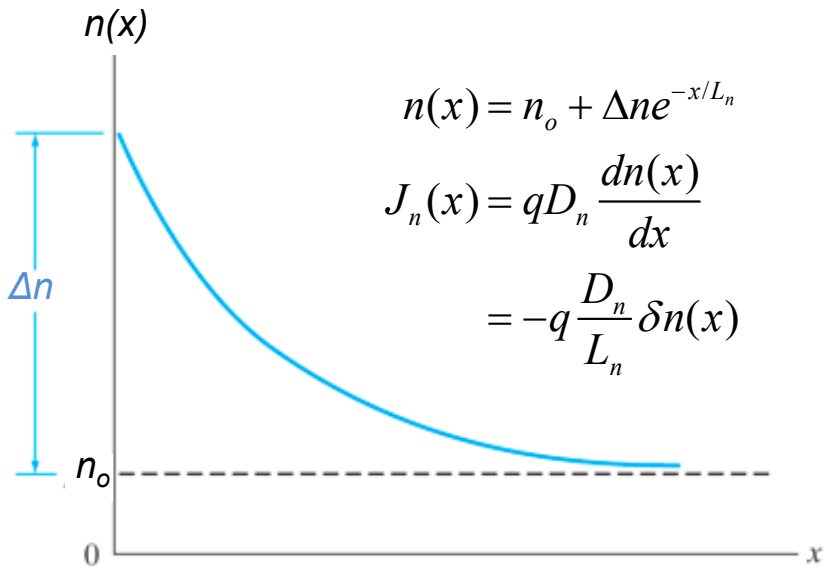
$$\delta n(x=0) = \Delta n = C_1 + C_2$$

$$\delta n(x=\infty) = 0 = C_1 e^{x/L_n}$$

$$C_1 = 0, \quad C_2 = \Delta n$$

$$\delta n(x) = \Delta n \, e^{-x/L_n}$$

Electron Injection at x=0



$$n(x) = n_o + \Delta n e^{-x/L_n}$$

$$J_n(x) = qD_n \frac{dn(x)}{dx} = -q\frac{D_n}{L_n}\delta n(x)$$

$$J_n(x) = qD_n \frac{dn}{dx} = qD_n \frac{d(n_o + \delta n)}{dx} = qD_n \frac{d\delta n}{dx}$$

$$= -q\frac{D_n}{L_n}\Delta n e^{-x/L_n} = -q\frac{D_n}{L_n}\delta n(x)$$

Diffusion Length

- The injected electrons or holes decay exponentially due to recombination
- The average distance an electron or hole travels before recombining is the diffusion length (we will prove this next)
- Mathematically, this is the distance where the quantity of injected carriers drops to $^1/_e$ (0.368) times the value at the point of injection

Diffusion Length: Statistical Interpretation

Probability A Hole Injected at x = 0 Survives to "x":

$$\frac{\delta p(x)}{\Delta p} = \frac{\text{Number of Excess Holes at "x"}}{\text{Number of Excess Holes at x=0}} = \frac{\Delta p e^{-x/L_p}}{\Delta p} = e^{-x/L_p}$$

Probability A Hole Will Recombine in the Interval dx :

$$\frac{\text{Holes Lost in Interval dx}}{\text{Number of Excess Holes at x}} = \frac{\delta p(x) - \delta p(x+dx)}{\delta p(x)} = \frac{-\left(d\delta p(x)/dx\right)dx}{\delta p(x)} = \frac{-\left(-\frac{1}{L_p}\right)\Delta p e^{-x/L_p}dx}{\Delta p e^{-x/L_p}} = \frac{1}{L_p}dx$$

The probability a hole will recombine at "x" equals the probablility it survives to "x" times the probability it recombines at "x":

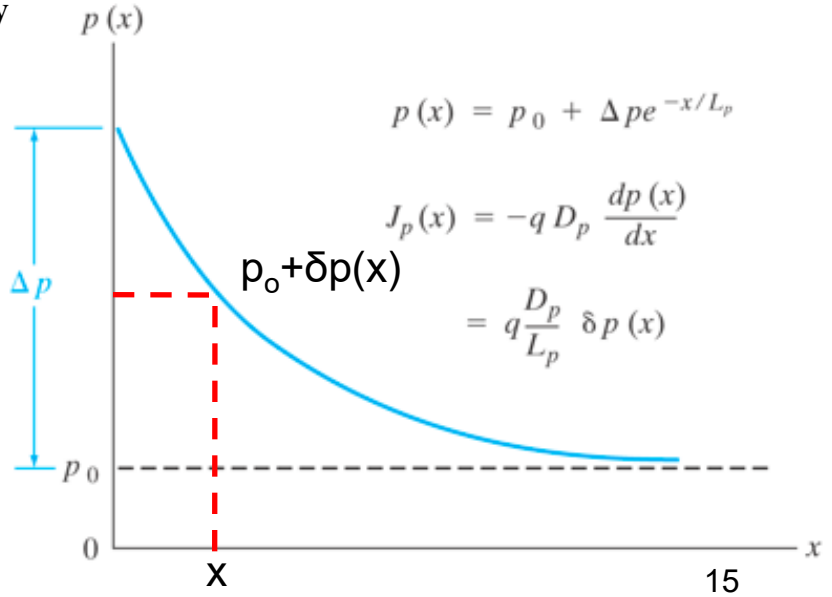
$$\text{P(hole recombination at "x")} = \left(e^{-x/L_p}\right)\left(\frac{1}{L_p}dx\right) = \frac{1}{L_p}e^{-x/L_p}dx$$

But recall the definition of the expectation value of a function:

$$\langle f(x)\rangle = \int_{-\infty}^{\infty} f(x)P(x)dx$$

so for the function $f(x) = x$:

$$\langle x\rangle = \int_0^{\infty} x\frac{1}{L_p}e^{-x/L_p}dx = L_p$$



Current Flow and Charge Loss

$$\delta p(x) = \Delta p \bullet e^{-x/L_p}$$

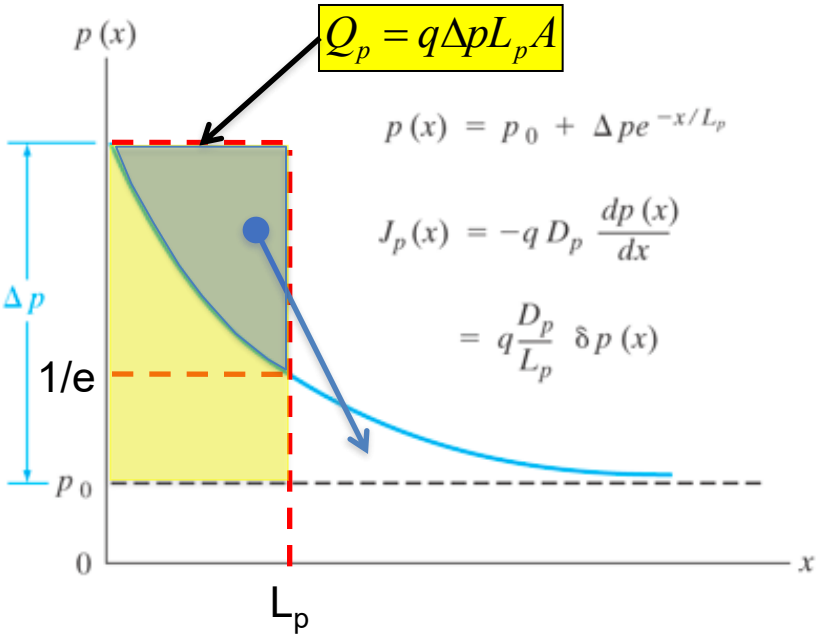
Note, the total number of excess carriers is:

$$A \times \int_0^\infty \Delta p e^{-x/L_p}\, dx = -A\Delta p L_p e^{-x/L_p}\Big|_0^\infty = -A\Delta p L_p \left(e^{-\infty/L_p} - e^0\right)$$

$$= -A\Delta p L_p \left(e^{-\infty/L_p} - e^0\right)$$

$$= A\Delta p L_p \quad \leftarrow \text{Total excess carrier}$$

$$\boxed{Q_p = q\Delta p L_p A} \quad \text{Total stored charge}$$



Hole current J_p is:

$$J_p(x) = -qD_p \frac{dp(x)}{dx} = -qD_p \frac{d\delta p(x)}{dx} = q\frac{D_p}{L_p}\delta p(x)$$

$$J_p(0) = q\frac{D_p}{L_p}\Delta p = q\left(\frac{L_p^2}{\tau_p}\right)\frac{1}{L_p}\Delta p = \frac{qL_p\Delta pA}{\tau_p A} = \frac{Q_p}{\tau_p A}$$

$$J_p(0) = \frac{Q_p}{\tau_p A} \quad , \quad I_p(0) = \frac{Q_p}{\tau_p}$$

- The current flow into the device at x=0 is equal to the charge loss due to recombination